

ONE LABEL, THREE NUMBERS

each reads as "time to depletion"

reserve-life ratio

reserves $\div$ production

arithmetic — not a forecast

trend index

a fitted trend vs its own minimum

an index — not a stock ratio

removals-only time

log margin $\div$ mortality

a pressure scale — not depletion

public data: G3P · USGS · RAM

THE TYPED LEDGER

each moiety in its own currency

biomass

funds

biodiversity

typed fluxes — never summed

conservation

from the incidence structure

positivity

outflows vanish with their donor

services

readouts, not conserved mass

three depletion times, not one

turnover · frozen rate · hitting time

NO WEIGHT HIDES A DEFICIT

the vector carries the certificate

weighted aggregate ≥ 0

for every weight ≥ 0



componentwise

one floor fails

double counting

no phantom mass

no weighting certifies all floors

*an aggregate ≥ 0 is not each floor ≥ 0
each claim carries its own predicate*

Typed Flux Ledgers and Depletion Arithmetic: Conservation, Componentwise Diagnostics, and the Semantics of Depletion Horizons